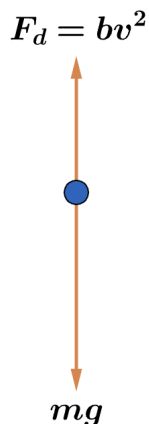


2024 F=ma Exam: Problem 24

Kevin S. Huang



For a particle falling under quadratic drag $F_d = bv^2$, recall the characteristic time τ determines how long it takes the particle's speed v to be on the order of the terminal velocity v_t . Similarly, the characteristic length l determines how far the particle has traveled when its speed $v \sim v_t$. These intrinsic parameters can be found by dimensional analysis:

$$\begin{aligned} [m] &= M \\ [g] &= \frac{L}{T^2} \\ [b] &= \frac{[F_d]}{[v]^2} = \frac{ML/T^2}{L^2/T^2} = \frac{M}{L} \end{aligned}$$

To make a length, we can divide m by b (g has dimensions of time so it can't appear). Thus,

$$l = \frac{m}{b}$$

In our case, we are given $b \propto A\rho_f \propto R^2\rho_f$ and we also have $m \propto R^3\rho_s$,

$$l \propto \frac{R^3\rho_s}{R^2\rho_f} = \frac{R\rho_s}{\rho_f}$$

The displacement when $v = v_t/2$ is proportional to l so the answer is B.